

Decoupling Safety Guarantees from Policy Competence in Multi-Robot Control Barrier Function Filters

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Abstract

Control barrier function (CBF) safety filters are widely used to wrap learned policies in multi-robot systems, on the premise that the filter makes the resulting system “safe.” We argue that this conflates two properties: a velocity-level constraint (the filter does not command a colliding action) and episode-level competence (the policy accomplishes the task). Using a three-robot ROS 2/Gazebo system with a soft CBF filter and 72 archived evaluation runs, we show empirically that the two come apart. Competence varied widely across these runs (area coverage from 1.1% to 98.6%, although runs differ in world and horizon), yet inter-agent contacts were absent in 69 of 72 runs and were confined to three swarm-size configurations in the two smallest arenas, so the collision count carried no information about competence. Within a 27-episode ablation in which coverage ranged from 24% to 52%, the wall-contact count was uncorrelated with coverage (Spearman $\rho = 0.04$, $p = 0.85$), and for one checkpoint it varied between 0 and 19 contacts across repeated evaluations. The filter is also soft: in one 3,500-step run the minimum inter-robot distance was 0.382 m, below the 0.45 m configured margin, with no contact. We propose a three-class taxonomy (competent-and-filtered, incompetent-and-filtered, unfiltered) and a reporting checklist: a competence metric, the minimum clearance against the configured margin, the collision definition relative to the filter’s own radius, and repeated evaluation. The evidence comes from one system with small samples, and we state its limits explicitly.

Keywords: control barrier functions, safe reinforcement learning, multi-robot systems, safety filtering, policy evaluation

1 Introduction

A control barrier function (CBF) renders a designated safe set forward invariant by solving, at each control step, a quadratic program (QP) that minimally perturbs a nominal control input subject to a barrier constraint [1]. In learning-enabled robotics, CBF filters are often proposed as a way to make an otherwise uncertified learned policy safe: the policy proposes an action, the filter passes it through or overrides it, and the composed system is expected to inherit a collision-avoidance property independent of the policy’s quality [2]. Recent work continues this pattern by enforcing CBF constraints during RL training rather than only at deployment [8], by combining CBFs with actor-critic architectures through Koopman representations [5], and by applying CBF-informed RL to connected and automated vehicles [7] and to mobile robot navigation under risk [3].

This literature is framed around a positive claim: the filter makes the system safe. We do not dispute that a correctly implemented filter bounds the commanded velocity near detected obstacles. We examine what that does *not* imply, using a multi-robot system for which the underlying data are archived and the code is public. Our contribution is narrow. We show that (i) a collision count is a poor proxy for competence, because it can be identical for policies whose task performance differs by a large factor; (ii) a soft CBF filter does not enforce its configured margin as a hard constraint, so “zero collisions” and “the margin was respected” are different statements; and (iii) collision counts in a simulator-in-the-loop harness can themselves vary between repeated evaluations of the same checkpoint. We propose a taxonomy and a reporting checklist in response.

2 Related Work

Ames et al. [1] establish CBFs as a general safety certification tool via forward invariance of a safe set. Cheng et al. [2] use a barrier-function filter as an online safety layer around a learned policy for safety-critical continuous control, which is similar in spirit to the architecture we examine; their emphasis, like most follow-on work, is on showing that the filter prevents constraint violation rather than on characterizing what the filtered system can still do when the policy is weak. CBF-RL [8] internalizes CBF constraints into training so that safety need not be enforced online; this is complementary to our setting, in which an online filter is retained. Koopman-based CBF filters for actor-critic RL [5] and CBF-informed RL for connected and automated vehicles [7] extend the filtering idea to new system classes. A risk-aware dynamic safety filter for mobile robot navigation [3] reasons about risk rather than binary safety. We did not find, in these works, a competence metric reported alongside the

safety metric, but we did not conduct a systematic review and do not claim that none exists. Henderson et al. [4] document how evaluation protocol and seed count affect conclusions in deep RL; our observation that collision counts vary between repeated evaluations of one checkpoint is a simulator-level instance of the same concern. Yu et al. [9] establish MAPPO with a centralized critic as a strong baseline for cooperative multi-agent tasks; we use a policy in this family only as one of the controllers under test.

3 System Under Test

Three TurtleBot3 Waffle robots (differential drive, 24-beam LiDAR) perform cooperative area coverage in Gazebo Harmonic under ROS 2 Jazzy, in five worlds (**cafe**, **warehouse**, **depot**, **office**, **maze**). Three controllers are evaluated: a Frontier Heuristic (frontier selection, A* routing, Voronoi partitioning, behavior-tree recovery), a Random Walk, and a MAPPO policy trained on **cafe**. All three issue a nominal command $(v_{\text{nom}}, \omega_{\text{nom}})$ to the same safety stack. Environment, controllers and training are described in a companion paper [6]; we summarize only what is needed here.

Pre-filter overrides. Before the CBF solve, the environment applies two fixed rules to every controller’s command. If the minimum LiDAR range over all beams is below 0.22 m and the commanded forward speed is positive, forward speed is replaced by -0.08 m/s. If another robot is within 0.45 m and within 72° of the heading, forward speed is replaced by -0.05 m/s.

Soft CBF filter. The filter solves, with OSQP,

$$\min_{v, \omega, \delta} (v - v_{\text{nom}})^2 + 0.05(\omega - \omega_{\text{nom}})^2 + P \delta^2 \text{ s.t. } A_k v + B_k \omega + \delta \geq -\gamma h_k \quad \forall k, \quad (1)$$

with $|v| \leq 0.22$ m/s, $|\omega| \leq 1.0$ rad/s and $0 \leq \delta \leq 10$. The barrier terms h_k come from LiDAR returns within 0.6 m, evaluated at two lookahead points $(\pm l, l = 0.12$ m) on the heading axis with obstacle radius $d_{\text{safe,obs}} = 0.20$ m, and from neighbors within 1.2 m with agent radius $d_{\text{safe,agent}} = 0.45$ m; $\gamma = 2.0$ and $P = 1000$. After the solve, forward speed is set to $\min(0, v)$ if any return in the forward cone (five beams) is below 0.24 m. If the QP is infeasible the filter returns $(\min(0, v_{\text{nom}}), \omega_{\text{nom}})$. Because of the slack variable δ , the barrier constraints may be violated at a price, so *the configured radii are soft*. The agent term uses a relative-distance barrier that treats the neighbor’s velocity as bounded and unmodeled rather than estimating it. We therefore make no formal safety claim for this filter; everything below is an empirical observation about its behaviour in simulation.

Collision metrics. A *wall contact* is counted when the minimum LiDAR range is below

0.14 m, the approximate sensor-to-bumper distance of the robot. An *agent contact* is counted when two robots are closer than 0.20 m after the first 15 steps of an episode. Note that the agent-contact threshold (0.20 m) is much smaller than the filter’s agent margin (0.45 m): the metric does not test whether the margin was respected.

4 Empirical Characterization

All numbers below come from files archived in the project repository (`checkpoints/`) and were recomputed from the raw per-run records. Each run is a single episode of a single configuration unless stated otherwise.

4.1 Safety audit across swarm size and world

Table 1 reports a robot-count sweep with the Frontier Heuristic, 200 steps per run, one run per cell.

Table 1: Frontier Heuristic, 200 steps, one run per cell. Entries are agent contacts / wall contacts.

World	$N=2$	$N=3$	$N=5$	$N=8$
cafe	0 / 0	0 / 0	2 / 42	8 / 16
warehouse	0 / 0	0 / 0	0 / 0	0 / 4
depot	0 / 0	0 / 0	0 / 73	0 / 31
office	0 / 0	0 / 8	0 / 14	2 / 0
maze	0 / 0	0 / 0	0 / 0	0 / 135

Agent contacts occurred in 3 of 20 configurations, all at $N \geq 5$ in the two smallest arenas (**cafe**, 17×8 m, and **office**), consistent with the fixed margins occupying a larger share of free space as density rises, though we did not test that explanation. Three longer single-controller runs (24,000-step **warehouse**, 3,500-step **depot**, 3,500-step **office**) recorded no agent contact. Across the 25-run controller matrix and the 27-episode ablation described below, no agent contact occurred. In total, 69 of the 72 archived sweep, matrix and ablation runs had no agent contact.

4.2 The margin is soft

In the 3,500-step **office** run the minimum inter-robot distance was 0.382 m, below the 0.45 m agent margin, with no agent contact (the contact threshold is 0.20 m). The **depot** run’s minimum was 0.552 m and the 24,000-step **warehouse** run’s was 0.814 m. This is the

behaviour expected of a soft constraint; it means that a zero-contact record, taken alone, does not show that the configured margin was maintained.

4.3 Dynamic hazards

Two scenarios were run once each in Gazebo with a scripted moving robot as the hazard, with the obstacle radius set to 0.22 m (the dynamic test script’s setting, which differs slightly from the 0.20 m used elsewhere). In a head-on approach at a closing speed of 0.38 m/s the filter shed forward speed and reversed, with minimum centre distance 0.349 m, minimum LiDAR range 0.211 m and no contact. In an orthogonal crossing at 0.22 m/s the robot yielded to standstill and then resumed, with minimum centre distance 0.437 m and no contact. Solver-level unit tests with synthetic LiDAR returns check that forward speed is shed as an approaching return closes (included in the repository’s CI). Each Gazebo scenario is a single run and the raw traces are not archived, so these are illustrations, not statistical evidence.

4.4 Collision counts do not track competence

The 25-run controller matrix (5 worlds $\times$ 5 controller/scenario conditions, one episode each, 150 steps) spans coverage from 1.2% to 57.0% and contains no agent contact. In **cafe** the Frontier Heuristic (57.0% coverage), Random Walk (22.6%) and nominal MAPPO (28.2%) all recorded zero wall and zero agent contacts. The 27-episode ablation re-evaluates three MAPPO checkpoints trained under different reward coefficients, three seeds with three repeated evaluations each (the repeated evaluations of one checkpoint are not independent samples; see the companion paper). Coverage ranged from 24.0% to 51.6%. Agent contacts were zero in all 27 episodes, and the wall-contact count was uncorrelated with coverage (Spearman $\rho = 0.04$, $p = 0.85$, $n = 27$; Pearson $r = -0.04$). The mean wall-contact count per episode was 4.00, 0.78 and 0.00 for the arms with mean coverage of 28.0%, 31.0% and 40.5%, but the arm difference is driven by a few episodes: for the Baseline checkpoint the count was 0 in 6 of 9 episodes and 6, 19 and 11 in the other three (all from one evaluation pass). Repeated evaluation of an identical checkpoint and seed argument therefore changes collision counts as well as coverage, which we attribute to real-time simulator timing rather than to policy sampling, since evaluation uses greedy actions.

The pattern matters for reporting. A collision count that is identically zero across policies of very different competence cannot rank them, and a count that varies between 0 and 19 across repeated evaluations of one checkpoint is not a stable property of the policy.

5 A Taxonomy and Reporting Checklist

We classify a filtered learned system along two axes: whether a safety filter is active, and policy competence under it (a task metric such as coverage rate or goal-reach rate).

- **Class A, competent and filtered.** The policy accomplishes the task and the filter rarely binds.
- **Class B, incompetent and filtered.** The policy performs poorly while the filter prevents constraint violations. A collision metric alone cannot distinguish this from Class A.
- **Class C, unfiltered.** Safety and competence are entangled and any low collision rate is attributable to the policy.

In our data, policies spanning a wide coverage range all fall under the same zero agent-contact record (Section 4.4); on the collision metric the weaker and stronger ones are indistinguishable. We therefore recommend that papers reporting collision rates for CBF-filtered learned policies also report:

1. a task-competence metric alongside the collision metric;
2. the minimum clearance achieved, against the configured margin;
3. the collision definition and its threshold relative to the filter’s own radius;
4. the filter’s intervention frequency and magnitude per controller (we did *not* measure this, which is a limitation of our own study); and
5. repeated evaluation of each checkpoint, with the unit of statistical independence stated.

6 Limitations

The evidence comes from one system, three controllers and a simulator whose evaluation is not deterministic: three evaluations of an identical checkpoint and seed produced different results in the companion study, and the single-episode cells in Table 1 and in the controller matrix inherit that variance. Sample sizes are small, so we report counts and one rank correlation rather than significance tests on the sweep or matrix. The dynamic-hazard results are single runs. The filter is soft and we offer no formal guarantee. We did not measure how often or by how much the filter and the pre-filter overrides modify each controller’s commands, so we cannot say how much of any controller’s behaviour they mediate. We do

not claim that CBF filters are ineffective: the filter and overrides operated as configured throughout, and a velocity-level bound is a useful property in its own right.

An earlier version of this paper reported a frozen, undertrained MAPPO policy with zero displacement and a higher wall-contact count than two baselines as its central example. We withdrew it: it came from an earlier filter implementation (a SciPy SLSQP solver and different collision thresholds), its raw records are not archived, and it cannot be reproduced with the current code. The thesis here rests instead on the archived runs above.

7 Conclusion

In a three-robot Gazebo system with a soft CBF filter, agent contacts were absent in 69 of 72 archived runs while coverage ranged from 1.1% to 98.6% (across worlds and horizons), wall-contact counts were uncorrelated with coverage within an ablation, and the configured agent margin was not strictly maintained. A collision count therefore cannot stand in for competence, and a zero-contact record does not show that a soft margin held. We propose reporting a competence metric, minimum clearance, the collision definition and repeated evaluations alongside any collision rate, and we note that our own study lacks the filter-intervention measurement that we recommend.

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